Clean Enough to Recycle? A Multi-Task CV Approach to Contamination-Aware Waste Sorting

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Computer Science

Natnael Tadesse [1] Kennesha Garg [1] Christian Roy [1]

[1]Department of Computer Science, Stanford University

Motivation

Waste sorting systems today classify items by material type. They do not assess whether an item is recyclable in its current state. Food residue, grease, and liquid stains can make an otherwise recyclable item non-recyclable.

The problem: A single contaminated container can cause an entire sorted bale to be rejected at a Materials Recovery Facility (MRF). Contamination state matters at every step of the sorting pipeline.

Our goal: We treat contamination-aware recyclability as a multi-task computer vision problem. We train a model to classify waste, detect contamination, and localize it at the pixel level.

- **Input:** 224×224 RGB image of a waste item or litter scene
- **Outputs:** waste category label, binary contamination flag, per-pixel contamination mask
- **Challenge:** no real contamination labels exist, so we generate synthetic supervision using procedural overlays

Figure 1. Both items are cardboard pizza boxes. Food residue is the only difference. **Left:** clean. **Right:** contaminated.

Datasets

- **ZeroWaste-f** (training): 4,661 frames from a real MRF conveyor belt. Four categories: rigid plastic, soft plastic, cardboard, metal. All images are clean.
- **TACO** (evaluation): approximately 1,500 litter images across 60 categories from streets, parks, and beaches.

Our annotation: We annotated **1,403 TACO images** using CVAT. All images received binary labels. All **100 contaminated examples** (7.1%) also received hand-drawn pixel masks. Held out entirely from training.

Figure 2. ZeroWaste-f conveyor belt frames (**left, center**) and CVAT annotation interface showing bounding boxes on a contaminated image (**right**).

References

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Model: Multi-Task ResNet-50

Shared Backbone

ResNet-50 pretrained on ImageNet, fine-tuned end-to-end. Outputs a $7\times7\times2048$ feature map shared across all three heads.

Head 1: Classification

Global average pooling $\to$ linear layer $\to$ 4-class logits.

$$\mathcal{L}_{\text{cls}} = -\sum_{c=1}^{4} y_c \log \hat{y}_c$$

Head 2: Contamination Presence

Global average pooling $\to$ scalar logit. Positive class weight $w{=}5.0$ penalizes false negatives to reflect MRF bale-rejection costs:

$$\mathcal{L}_{\text{pres}} = -\left[w \cdot y \log \hat{y} + (1-y)\log(1-\hat{y})\right]$$

Head 3: Contamination Localization

Five-stage bilinear upsampling decoder ($2048\to512\to256\to128\to64\to32$), outputs a 224×224 mask.

$$\mathcal{L}_{\text{loc}} = \mathcal{L}_{\text{BCE}} + \mathcal{L}_{\text{Dice}}, \qquad \mathcal{L}_{\text{Dice}} = 1 - \frac{2\sum_i p_i g_i}{\sum_i p_i + \sum_i g_i}$$

Combined Loss

$$\mathcal{L} = \lambda_1\mathcal{L}_{\text{cls}} + \lambda_2\mathcal{L}_{\text{pres}} + \lambda_3\mathcal{L}_{\text{loc}}, \quad \lambda_1{=}1.0,\ \lambda_2{=}2.0,\ \lambda_3{=}1.0$$

AdamW, lr $=10^{-4}$, weight decay $=0.05$, cosine annealing with 1-epoch warmup.

Synthetic Contamination Pipeline

ZeroWaste-f has no contamination labels, so we composite synthetic overlays onto clean training images at runtime. We simulate three types: grease, food residue, and liquid spills. Each is a semi-transparent Gaussian blob. We place 2 to 6 blobs per image with opacity drawn from $[0.55, 0.95]$. Any pixel with opacity above 0.25 is labeled contaminated for free.

Results

Condition	Synth	cls acc	pres P	pres R	pres F1	mIoU
B0 (cls only)	0%	60.0%	–	–	–	–
B1 (all heads)	0%	–	0.000	0.000	0.000	0.000
B1.5 (all heads)	50%	–	0.081	**0.480**	**0.138**	0.017
B2 (all heads)	100%	–	0.064	0.180	0.094	0.016
B2 (synth val)	100%	63.0%	–	–	1.000	0.759

Table 1. B1–B2 evaluated on real TACO. B2 (synth val) on synthetic held-out only, shown to quantify the sim-to-real gap. Random baseline = 7.1%.

Ablation and Precision-Recall

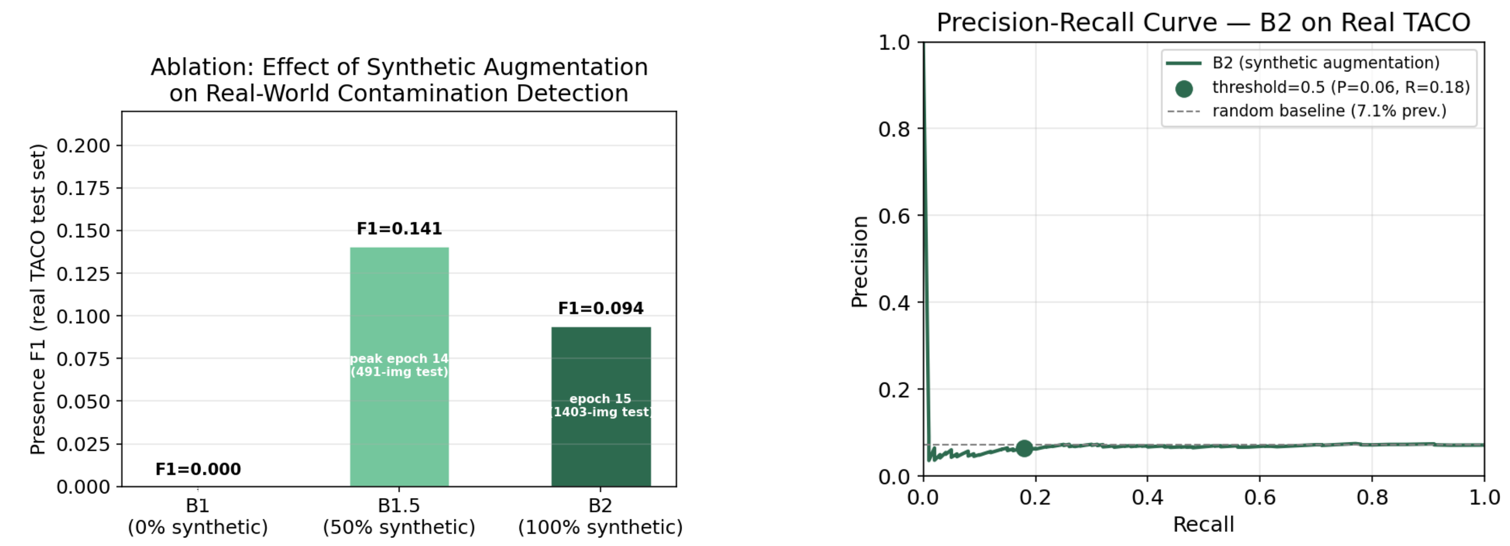

Figure 3. **Left:** Peak presence F1 on real TACO across ablation conditions. **Right:** Precision-recall curve for B2. Precision stays near the 7.1% random baseline across almost the full recall range.

Grad-CAM Analysis

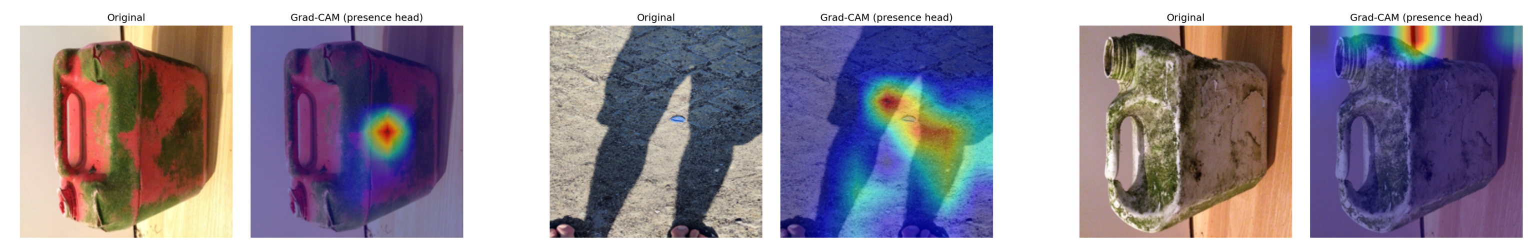

Figure 4. **Left:** Success – activation centered on algae covering a container. **Center:** Failure – background saliency; model attends to a person's legs. **Right:** Failure – texture confusion; sandy surface attracts activation.

Key Findings

- **Synthetic supervision is necessary.** Without it (B1), the presence head predicts every image as clean (F1 = 0.000).
- **Synthetic training works in-distribution.** B2 reaches F1 = 1.000 and mIoU = 0.759 on the synthetic validation set.
- **The sim-to-real gap is large.** B2 scores F1 = 0.094 and mIoU = 0.016 on real TACO. The model learned to detect blob artifacts rather than contamination as a concept.
- **Three Grad-CAM failure modes:** background saliency, texture confusion, and object-correct but region-wrong localization.
- **B1.5 outperforms B2 on real data** (F1 = 0.138 vs. 0.094).

Future Work

- Apply domain adaptation between synthetic and real contamination distributions to close the sim-to-real gap
- Use SAM 3 text-prompted segmentation on unlabeled TACO images to generate real contamination masks at no annotation cost
- Crowdsource more contamination labels so the model can learn from real examples directly